



Memory dual N-back training using mobile devices and its effect on working memory capacity

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Abstract

Working Memory (WM) considers the ability to process and store information and is commonly translated into the acquisition of new knowledge and skills, e.g., in the form of a memory strategy, such as an information grouping strategy. Therefore, increasing the efficiency of WM, particularly through non-invasive methods, such as training, becomes one of the crucial issues in enhancing one's cognitive functioning. Considering the average mobile device usage time, smartphones can be considered up-to-date training tools. Cognitive training using mobile devices can contribute to improvements in the field of online learning by providing a tool for improving students' cognitive skills. This study investigated the effectiveness of memory training using a mobile device, i.e., a smartphone. The training was assessed in a sample of young adults ($N=59$) within a longitudinal design involving a pre-test and double post-test with a training and control group. The Dual N-back training improved WM capacity and boosted important proximal and distal outcomes. Overall, our study indicated an increase in WM as a result of the training using smartphones, thus providing important cues for enhancing WM capacity by use of digital training on mobile devices on a daily basis.

Keywords Working memory · Working memory training · Dual N-back training · Digital span task · Dual-task training

Introduction

The concept of Working Memory (WM) describes the ability to simultaneously store and process information necessary for goal achievement (Baddeley, 2000). Based on recent research (Cowan, 2013), two WM components have been distinguished: the range and the control of attention. The range of attention concerns the amount of information that can be actively preserved at a given time, whereas the control of attention concerns the ability to focus only on relevant information and ignore irrelevant information. WM is considered a cognitive system responsible for the short-term storage of information, as well as for its processing, supervision, and coordination (Oberauer et al., 2003), thus is recognized as an important predictor of a wide range of cognitive abilities (von Bastian et al., 2013). WM is pointed out as having positive implications

for everyday functioning, resulting in, e.g., lower inattention (Spencer-Smith & Klingberg, 2016), reasoning ability in older adults (Klimova, 2016), verbal learning and cognitive control (Fisher et al., 2009), and creativity (Gong et al., 2023). WM capacity is also considered one of the best predictors of intelligence (Au et al., 2015; Jaeggi et al., 2008). Existing research depicted poor WM as associated with learning difficulties and lower academic performance, irrespective of general intellectual abilities (Dentz et al., 2020). Moreover, poor WM performance characterizes neurodevelopmental disorders, such as attention deficit disorder (ADD) and learning disability (Minear et al., 2016). WM capacity allows prediction of performance on complex cognitive tasks, i.e., reading and language comprehension and mathematical performance (Dentz et al., 2020). Therefore, an activity aimed at maintaining or enhancing the optimal functioning of WM becomes extremely important. Drawing from that, this study aimed to test whether systematic WM training improves WM functioning regarding capacity understood as the amount of stored information.

People are very eager to use various forms of cognitive enhancement and try to improve the efficiency of their minds, hence the stated goal is very current. Since cognitive

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